# Cross-Architecture Transferability is Width-Confounded: Diagnosis and Subspace Correction

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## Abstract

Selecting a pretrained encoder without target labels is a standard preprocessing step in transfer learning. RankMe, a widely used label-free score, was validated primarily within architecture families. We show that its effective-rank score is strongly confounded by penultimate-layer width when heterogeneous architecture families are compared: under an isotropic random-feature model, RankMe effective rank scales as $\Theta(d)$, and empirically its cross-family rankings track width rather than transfer quality. We propose Subspace-Corrected Ranking (SCR), which augments target spectral information with the relative geometry between each candidate's target features and its features on a fixed reference dataset. Its parameter-free member $\mathrm{SSR}^{\ddagger}$ subtracts log subspace alignment from log effective rank. On a common support of 22 pool–task units from MV, VTAB, and EX, $\mathrm{SSR}^{\ddagger}$ reduces mean regret from $+0.169$ (RankMe) to $+0.066$; the paired difference is $-0.1025$ (target-name-clustered 95% CI $[-0.164, -0.039]$, $p = 0.00114$). $\mathrm{SSR}^{\ddagger}$ is intended for cross-family selection, not as a universal replacement for RankMe: RankMe remains preferable within a family. A five-task MV experiment further shows that $\mathrm{SSR}^{\ddagger} \rightarrow$RankMe can retain similar regret while reducing forward evaluations, which we treat as a proof of concept rather than a general cost guarantee.

## 1 Introduction

Transfer learning starts with a choice: which pretrained encoder to fine-tune on the target task. When candidate encoders share the same architecture but differ in pretraining recipe, the choice can be made cheaply via a label-free score on the encoder's penultimate features. The dominant such score, *RankMe* [8], computes the effective rank $r_{\mathrm{eff}}(H)$ of the feature matrix and ranks candidates by $\log r_{\mathrm{eff}}$. In the within-architecture setting where RankMe was validated, the penultimate-layer dimension $d$ is constant across candidates and the score reflects representational quality.

The realistic deployment setting is different. A practitioner faced with the timm [30] or torchvision model zoo chooses between fundamentally different backbones—ResNet [10], ViT [5], ConvNeXt [16], EfficientNet [26], and others—whose penultimate dimensions vary by an order of magnitude. RankMe was validated on within-family variants [8]; however, practitioners selecting among heterogeneous backbones on model hubs such as timm [30] and Hugging Face Hub naturally apply it pool-wide, without the within-family stratification its validation assumed. In this cross-architecture setting, we show that RankMe-based selection fails systematically. The rankings it produces are dominated by the penultimate-layer width: on a pool of 132 pretrained variants spanning 35 multi-variant architectures, RankMe selects the widest available variant on every task, yielding regret worse than uniform random selection. The same width confound is observed on a 25-architecture VTAB pool,

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an 8-encoder SSL/multimodal pool, and a 51-variant EX pool spanning 10 tasks. In all four pools, label-free cross-architecture transferability estimation as currently practiced is unsound.

The mechanism is a consequence of RankMe's target-only spectral form under feature isotropy. A corrected Marchenko–Pastur argument shows that RankMe effective rank scales as $\Theta(d)$ at fixed positive aspect ratio, so $\log r_{\text{eff}}$ equals $\log d$ plus an aspect-ratio-dependent constant asymptotically. This formal statement is specific to RankMe; rotation invariance alone is insufficient to conclude that an arbitrary score is monotone in $d$.

The correction is relational rather than target-spectral. We propose *Subspace-Corrected Ranking* (SCR), which compares each candidate encoder's target features with reference-dataset features extracted by that same encoder. The score is invariant to a common change of representation basis, but unlike RankMe it is not determined by the target singular spectrum alone. Across the candidate pool, $\text{SA}_k$ correlates strongly negatively ($\rho_{\text{Sp}} = -0.964$) with the fraction of class-mean variance outside the reference subspace. This is mechanistic evidence, not a formal guarantee of downstream accuracy. The default member, $\text{SSR}^{\ddagger} = [\log r_{\text{eff}} - \log \text{SA}_k]/\log C$, is designed for cross-family selection; RankMe is retained for within-family variant selection.

**Contributions.** **(i) Diagnosis.** We establish RankMe's $\Theta(d)$ scaling under an isotropic random-feature model and document strong width confounding across heterogeneous model pools. **(ii) Correction.** We propose SCR, a family of label-free relational scores that incorporate within-candidate target–reference subspace geometry, with empirical mechanistic support on 126 architecture–task pairs. **(iii) Scope-aware pipeline.** We use $\text{SSR}^{\ddagger}$ for cross-family selection and RankMe within a family; the resulting five-task MV cost–quality result is a proof of concept. **(iv) Benchmark.** Four reproducible evaluation pools covering 52 task units across supervised and SSL/multimodal pretraining paradigms, with backbone widths spanning $d \in [192, 4096]$. The pools are deliberately calibrated at three distinct $d_{\max}/d_{\min}$ ratios ($5.3\times$, $10.7\times$, $21.3\times$), exposing the regimes in which existing corrections succeed or fail and providing a diagnostic testbed for future label-free transferability scores. Pre-computed features, all result tables, and scoring code are released for community use.

## 2 Related Work

**Transferability scores.** RankMe [8] ranks pretrained encoders by the effective rank of penultimate features, using $p_i = \sigma_i / \sum_j \sigma_j$; our focus is the heterogeneous-pool setting beyond its original within-family validation. VNE [13] uses the entropy of normalised squared singular values, but we do not extend our RankMe-specific theorem to VNE. Official classification DiSCO [33] uses target ground-truth labels to construct class-conditional quantities and is therefore reported as a label-dependent comparator, not a zero-label baseline. Other label-requiring scores include LogME [31], H-score [2], NCTI [29], LEEP [19], N-LEEP [15], PACTran [4], OTCE [27], GBC [22], TransRate [12], NCE [28], and Task2Vec [1]. Transfer learning more broadly is surveyed in Pan and Yang [21].

**Subspace alignment and random matrix theory.** Subspace alignment [7] was introduced for unsupervised domain adaptation via principal-angle geometry on the Grassmann manifold [9]. Neural network representation similarity via CKA and SVCCA [14, 24] uses related geometric ideas to compare layers; we use it differently here, as a label-free measure of generic-vs-specific feature structure relative to a fixed reference dataset encoded by each candidate. The Marchenko–Pastur law [18] has been applied to characterise weight matrices in deep networks [17]; we apply it directly to feature matrices to derive the width-confound mechanism.

**Two-stage and budgeted model selection.** Staged screening followed by precise evaluation is a standard design in neural architecture search and Bayesian hyper-parameter optimisation. In source-free transferability estimation [29, 31], the same logic supports a cross-architecture stage at low cost followed by a within-family stage; our contribution is the formal cost–quality analysis of such pipelines and the mixed-budget variant in which a small label budget is allocated only to the architecture-selection stage.

# 3 Setup

**Pools.** We use four architecturally distinct pools whose descriptions are summarised in Table 1 (full per-pool details in Appendices B–E). **MV pool** (multi-variant pool; 132 timm [30] variants, 35 multi-variant architectures, 5 tasks) is the only pool with multiple variants per architecture and is therefore the only pool on which two-stage pipelines are evaluated. **VTAB pool** [32] (25 architectures, 8 tasks) is single-variant and broader in target diversity. **SSL pool** (8 SSL/multimodal encoders including CLIP [23], DINOv2 [20], DINO [3], and MoCo [11]; 29 tasks) uses pretraining-objective diversity on a smaller set of backbones. **EX pool** (extended supervised pool; 51 timm variants from 35 canonical architectures including VGG-16, $d$=4096; 10 tasks: CIFAR-10/100, DTD, EuroSAT, Fashion-MNIST, FGVC-Aircraft, Flowers-102, Food-101, Oxford-Pets, SVHN) covers the same $d_{\max}/d_{\min}$=21.3× width span as MV pool with a single canonical variant per architecture and broader task coverage; all variants are ImageNet-1k supervised, features extracted on the full dataset. Across the four pools we cover 52 task units and architectures with penultimate dimensions $d \in [192, 4096]$.

Table 1: Four architecturally distinct evaluation pools used throughout the paper. The $d_{\max}/d_{\min}$ spread drives the severity of the width confound and the choice of correction (see §7).

| Pool | Archs | Variants | Tasks | $d$ range | $d_{\max}/d_{\min}$ |
|---|---|---|---|---|---|
| MV pool (multi-variant) | 35 | 119 (multi-var.) / 132 (full) | 5 | [192, 4096] | 21.3× |
| VTAB pool | 25 | 25 | 8 | [192, 2048] | 10.7× |
| SSL/multimodal pool | 8 | 8 | 29 | [384, 2048] | 5.3× |
| EX pool (supv.) | 35 | 51 | 10 | [192, 4096] | 21.3× |

**Scores.** Effective rank $r_{\text{eff}}$ follows RankMe [8]: $p_i = \sigma_i / \sum_j \sigma_j$ and $r_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$. For each candidate encoder $m$, $\text{SA}_k$ compares its target features with its STL-10 features; both are produced by encoder $m$ and therefore share dimension $d_m$. We use uncentred feature matrices, default $k = 20$, and clip positive quantities below at $10^{-12}$ before taking logarithms. Different candidates' subspaces are never compared directly. Official classification DiSCO and the other label-requiring methods are reported separately from the zero-label RankMe–$\text{SSR}^{\ddagger}$ comparison.

**Linear-probe ground truth.** We use the held-out accuracy (lp_acc) of $\ell_2$-regularised logistic regression with $C$=1 on a single 80/20 split. Strategy quality is reported as *regret*, the gap to the per-task oracle accuracy. Confidence intervals are obtained by task-level bootstrap with $n_{\text{boot}}$=10 000 resamples; paired bootstrap for head-to-head comparisons.

# 4 The Width Confound

We isolate the failure mechanism with two formal statements. Both proofs are in Appendix A.

**Lemma 1** (Marchenko–Pastur scaling)**.** *Let $H \in \mathbb{R}^{n \times d}$ have i.i.d. rows from $\mathcal{N}(0, \sigma^2 I_d)$. As $n, d \to \infty$ with $\gamma = n/d$ fixed, $r_{\text{eff}}(H)/d \to c(\gamma) > 0$ almost surely, where $c(\gamma)$ depends only on the aspect ratio. In particular $r_{\text{eff}}(H) = \Theta(d)$.*

**Proposition 2** ($\text{SSR}^{\ddagger}$ is relational and jointly basis-invariant)**.** *For candidate encoder $m$ and $Q \in O(d_m)$, jointly replacing $(H_{m,T}, H_{m,R})$ by $(H_{m,T}Q, H_{m,R}Q)$ leaves their principal angles, $\text{SA}_k$, and $\text{SSR}^{\ddagger}$ unchanged. Nevertheless, $\text{SSR}^{\ddagger}$ is not determined by the singular values of $H_{m,T}$ alone: it additionally depends on the relative geometry of the target and reference right-singular subspaces.*

Lemma 1 is a RankMe-specific statement; rotation invariance alone does not imply monotonicity in $d$ for an arbitrary score. Proposition 2 explains how $\text{SSR}^{\ddagger}$ adds information absent from the target spectrum while remaining independent of an arbitrary common basis choice. Empirically, on EX the mean Kendall correlation between RankMe and $d$ is $\tau = +0.617$ across 10 tasks, confirming strong width concordance.

**Empirical confound across pools.** On MV pool the per-task Spearman correlation $\rho_{\text{Sp}}(\text{RankMe}, d)$ has mean $+0.61$ across the 5 tasks; on the VTAB pool the mean is $+0.882$ (Table 2 uses full-dataset VTAB features; the cross-pool aggregation in Table 7 uses $n$=1000 subsampled features for $\text{SA}_k$ computation consistency); on the SSL pool the mean is $+0.815$. Crucially, $d$ itself does *not* predict linear-probe accuracy: $\rho_{\text{Sp}}(d, \text{lp_acc})$ is $-0.10$ on MV pool, $-0.45$ on the SSL pool, and

nominally $+0.20$ on VTAB—elevated by SVHN (a digit-recognition task where capacity genuinely helps, $\rho{=}+0.856$); excluding STL-10 (reference task) and SVHN, the VTAB median falls to $+0.02$, consistent with the other pools. Across all three pools, $d$ acts as a pure confound rather than a quality signal. Flat-RankMe regret exceeds uniform random regret in all four pools (Table 2; EX pool: Appendix E). On MV pool, RankMe selects `vgg16` ($d{=}4096$) on every one of the 5 tasks, yielding regret $0.125$ LP points worse than random. MV pool's CNN-only sub-pool reproduces this behaviour (`vgg16` chosen on $4/5$ tasks, Appendix B), confirming that the confound is not a ViT-vs-CNN artefact.

Table 2: Width confound across MV, VTAB, and SSL pools (EX pool diagnosis data in Appendix E). RankMe correlates strongly with $d$ (large positive $\rho_{\mathrm{Sp}}$) yet $d$ itself does not predict quality (near-zero or negative $\rho_{\mathrm{Sp}}(d, \mathrm{lp_acc})$), confirming that $d$ is a confound rather than a signal.

| Pool | Arch., tasks | $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ | $\rho_{\mathrm{Sp}}(d, \mathrm{lp_acc})$ | RankMe regret | Random regret |
|---|---|---|---|---|---|
| MV pool | 35, 5 | $+0.61$ | $-0.10$ | $+0.319$ | $+0.194$ |
| VTAB | 25, 8 | $+0.882$ | $+0.20$ | $+0.083$ | $+0.064$ |
| SSL | 8, 29 | $+0.815$ | $-0.45$ | $+0.103$ | $+0.066$ |

**VTAB per-task breakdown.** On the VTAB pool the width confound is equally systematic: Table 3 shows that Flat-RankMe selects the highest-$d$ available architecture on *all* 8 VTAB tasks without exception. Despite the absence of VGG-class encoders (maximum $d{=}2048$), the per-task Spearman correlation $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ ranges from $+0.838$ to $+0.915$, and per-task regret ranges from $+0.020$ (Fashion-MNIST, near-uniform pixel data where feature quality matters little) to $+0.200$ (Flowers-102, fine-grained texture).

Table 3: VTAB per-task width confound. Flat-RankMe regret $+0.083$ (mean), suboptimal on all $8/8$ tasks; $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ is large and positive on every task.

| Task | $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ | Regret | RankMe pick |
|---|---|---|---|
| CIFAR-10 | $+0.874$ | $+0.050$ | efficientnet_b3 ($d{=}1536$) |
| CIFAR-100 | $+0.869$ | $+0.155$ | efficientnet_b3 ($d{=}1536$) |
| DTD | $+0.877$ | $+0.085$ | resnet101 ($d{=}2048$) |
| Fashion-MNIST | $+0.886$ | $+0.020$ | wide_resnet50_2 ($d{=}2048$) |
| Flowers-102 | $+0.838$ | $+0.200$ | efficientnet_b3 ($d{=}1536$) |
| Pets | $+0.902$ | $+0.030$ | mobilenetv3_large ($d{=}1280$) |
| STL-10 | $+0.892$ | $+0.025$ | efficientnet_b2 ($d{=}1408$) |
| SVHN | $+0.915$ | $+0.100$ | efficientnet_b3 ($d{=}1536$) |
| **Mean** | $\mathbf{+0.882}$ | $\mathbf{+0.083}$ | — |

**Severity is driven by inter-architecture width variance.** The MV pool failure is sharpest because the pool spans $d \in [192, 4096]$, with VGG-16 occupying the high-$d$ tail. Removing the three VGG variants ($\sim 3\%$ of the pool) shifts Flat-RankMe regret by $-0.175$ LP, bringing it close to random (Table 6); the same removal does not change $\mathrm{SSR}^{\ddagger}$'s behaviour because subspace alignment is approximately invariant to which architectures occupy the wide-feature tail. This robustness contrast generalises: rotation-invariant scorers (RankMe, LogME, H-score) all collapse onto a single architecture when a wide outlier is present, while $\mathrm{SSR}^{\ddagger}$ and DiSCO continue to choose task-appropriate variants. At a per-task level (Table 4), Flat-RankMe selects `vgg16` on every MV pool task while $\mathrm{SSR}^{\ddagger}$ selects the task-appropriate variant within $0.07$ LP of the oracle on $4/5$ tasks.

**An independent confirmation via spectral truncation.** Recomputing $r_{\mathrm{eff}}$ on the top-$D$ truncated singular spectrum of each candidate provides an independent check that the confound lives in the singular-value tail. At $D{=}256$ on MV pool, the per-task Spearman correlation $\rho_{\mathrm{Sp}}(r_{\mathrm{eff}, D=256}, d)$ drops from $+0.63$ (full spectrum) to $+0.06$, and Flat-RankMe regret drops from $+0.319$ to $+0.094$, comparable to $\mathrm{SSR}^{\ddagger}$. Truncation is not a viable correction for two-stage use, however: small-$d$ architectures (ViT-tiny, DeiT-tiny, both $d{=}192$) are not truncated at all and retain the width ordering; the corresponding stage-1 regret on the multi-variant subset is $+0.204$, more than twice that of $\mathrm{SSR}^{\ddagger}$ (Appendix E).

Table 4: Per-task breakdown of Flat-RankMe and Flat-$\mathrm{SSR}^{\ddagger}$ on MV pool's multi-variant subset. RankMe selects `vgg16` on every task; $\mathrm{SSR}^{\ddagger}$ recovers task-appropriate architectures.

| Task | Oracle | Random | Flat-RM | Flat-$\mathrm{SSR}^{\ddagger}$ | RM picks | $\mathrm{SSR}^{\ddagger}$ picks | Oracle arch |
|---|---|---|---|---|---|---|---|
| CIFAR-10 | 0.970 | 0.785 | 0.550 | 0.870 | vgg16 | resnet50 | convnext_base |
| CIFAR-100 | 0.825 | 0.484 | 0.180 | 0.670 | vgg16 | resnet152 | convnext_base |
| DTD | 0.740 | 0.564 | 0.445 | 0.720 | vgg16 | convnext_base | convnext_base |
| EuroSAT | 0.955 | 0.881 | 0.765 | 0.905 | vgg16 | efficientnet_b3 | densenet201 |
| SVHN | 0.615 | 0.430 | 0.570 | 0.450 | vgg16 | efficientnet_b3 | densenet169 |

**Related empirical failures.** On the MV pool, LogME and the trace form of H-score also select `vgg16` on every task and degrade when the VGG variants are present (Table 5), whereas LEEP and NCTI do not show the same pattern. We report this only as an empirical observation: our formal width-confounding result is specific to RankMe under the stated isotropic model and does not cover arbitrary spectral or rotation-invariant scores.

Table 5: MV pool SOTA flat comparison (132v×5t). Columns report per-task mean regret with and without the three VGG variants in the pool. Feature-spectrum scores (RankMe, LogME, H-score) all select `vgg16` when present and degrade similarly; class-distribution scores (LEEP, NCTI) and the proposed $\mathrm{SSR}^{\ddagger}$ are robust.

| Method | With VGG | Without VGG | Δ | Type |
|---|---|---|---|---|
| Random | +0.194 | +0.193 | −0.001 | — |
| Flat-RankMe | +0.319 | +0.144 | −0.175 | label-free |
| Flat-LogME | +0.319 | +0.550 | +0.231 | label-based |
| Flat-H-score | +0.319 | +0.470 | +0.151 | label-based |
| Flat-NCTI | +0.162 | +0.155 | −0.007 | label-based |
| Flat-DiSCO | +0.108 | +0.108 | 0.000 | label-based |
| **Flat-**$\mathrm{SSR}^{\ddagger}$ | **+0.098** | **+0.098** | **0.000** | **label-free** |
| Flat-LEEP | +0.038 | +0.038 | 0.000 | label-based |

Table 6: Sensitivity of flat strategies to the presence of VGG-class features ($d$=4096) in MV pool. Columns report mean LP across the 5 tasks. RankMe and the trace-spectrum label-based scores are dominated by VGG when present; $\mathrm{SSR}^{\ddagger}$, DiSCO, and class-distribution scores (LEEP, NCTI) are robust.

| Method | With VGG | Without VGG | Δ | Type |
|---|---|---|---|---|
| Random | 0.629 | 0.630 | +0.001 | — |
| Flat-RankMe | 0.502 | 0.677 | +0.175 | label-free |
| Flat-NormRankMe ($-\log d$) | 0.600 | 0.449 | −0.151 | label-free |
| **Flat-**$\mathrm{SSR}^{\ddagger}$ | **0.723** | **0.723** | **+0.000** | label-free |
| Flat-DiSCO | 0.713 | 0.713 | +0.000 | label-free |
| Flat-LogME | 0.502 | 0.271 | −0.231 | label-based |
| Flat-H-score | 0.502 | 0.351 | −0.151 | label-based |
| Flat-LEEP | 0.783 | 0.783 | +0.000 | label-based |
| Flat-NCTI | 0.659 | 0.666 | +0.007 | label-based |

# 5 Subspace-Corrected Ranking

**Geometric premise.** For candidate encoder $m$, let $V_{m,\mathrm{ref},1:k}$ be the top-$k$ right singular vectors of its features $H_{m,D_{\mathrm{ref}}}$ on the fixed reference dataset, and write $P = V_{m,\mathrm{ref},1:k} V_{m,\mathrm{ref},1:k}^{\top}$, $Q = I - P$. Define

$$f_{\mathrm{spe}}(H, H_{\mathrm{ref}}) := \frac{\mathrm{tr}(Q\Sigma_B Q)}{\mathrm{tr}(\Sigma_B)},$$

the fraction of between-class covariance lying outside the reference subspace. $f_{\mathrm{spe}}$ measures how much of a candidate's discriminative variance is architecture-specific rather than shared with a generic

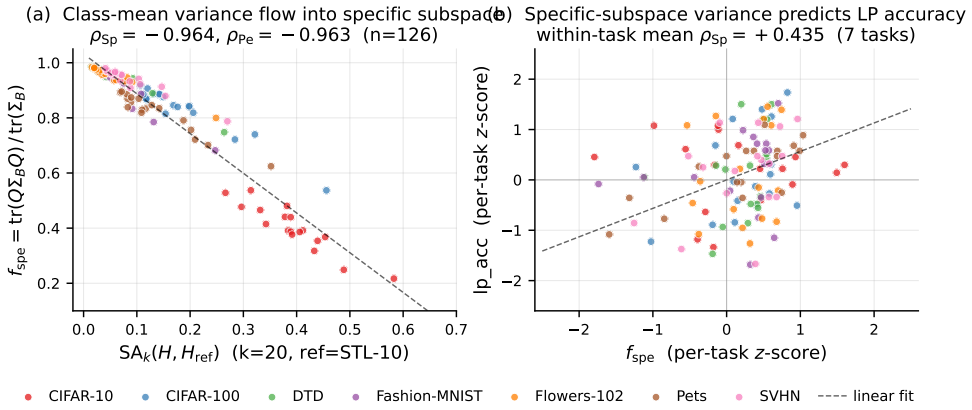



Figure 1: (a) Subspace alignment $\mathrm{SA}_k$ between a candidate encoder and the reference (STL-10) is nearly perfectly anti-correlated with the fraction of class-mean variance in the architecture-specific subspace, $f_{\mathrm{spe}}$. (b) Within-task, $f_{\mathrm{spe}}$ predicts linear-probe accuracy.

encoder. On the VTAB pool (126 architecture–task pairs), the subspace alignment $\mathrm{SA}_k(H, H_{\mathrm{ref}})$ is nearly perfectly anti-correlated with $f_{\mathrm{spe}}$ ($\rho_{\mathrm{Sp}} = -0.964$, all 7 tasks $\rho < -0.86$, Figure 1). Within each task, $f_{\mathrm{spe}}$ predicts LP accuracy with mean Spearman $\rho = +0.435$. Subtracting subspace alignment is therefore a label-free way to penalise candidates whose discriminative variance is absorbed by the generic subspace. The analysis covers 126 of the $25{\times}7{=}175$ non-reference architecture–task pairs: `resnet152` has a numerical $\mathrm{SA}_k$ failure (rank-deficient under $n{=}1000$ sampling), leaving 168 valid $\mathrm{SA}_k$ pairs; $f_{\mathrm{spe}}$ further requires full between-class covariance matrices, available for 18 of the 25 architectures ($126 = 18{\times}7$ pairs; see `results/vtab_fspe_analysis.csv`).

**Alignment and algorithm.** For candidate $m$, let the uncentred feature matrices be $H_{m,T} \in \mathbb{R}^{n_T \times d_m}$ and $H_{m,R} \in \mathbb{R}^{n_R \times d_m}$, both produced by encoder $m$. Let $V_T, V_R \in \mathbb{R}^{d_m \times k_m}$ contain their leading right-singular vectors, where $k_m = \min\{k, n_T, n_R, d_m\}$ and $k = 20$ by default. With principal angles $\theta_1, \ldots, \theta_{k_m}$,

$$\mathrm{SA}_k(H_{m,T}, H_{m,R}) = \frac{1}{k_m}\|V_T^\top V_R\|_F^2 = \frac{1}{k_m}\sum_{i=1}^{k_m}\cos^2\theta_i.$$

The complete computation is: extract target and reference features with the same candidate; compute their uncentred SVDs; retain the two top-$k_m$ bases; compute $\mathrm{SA}_k$ from their principal angles; compute RankMe $r_{\mathrm{eff}}$ from the target singular values; clip $r_{\mathrm{eff}}$ and $\mathrm{SA}_k$ below at $10^{-12}$; and return the scalar score below. Candidate feature dimensions may differ, because no cross-candidate subspaces are compared.

**Method family.** Subspace-Corrected Ranking is the family

$$\mathrm{SCR}(H, T; \beta, g) := \frac{\log r_{\mathrm{eff}}(H) + \beta \cdot g\big(\mathrm{SA}_k(H, H_{\mathrm{ref}})\big)}{\log C(T)},$$

where $g$ is a monotonically decreasing function and $\beta \geq 0$. Choosing $g(s) = -\log s$ with $\beta = 1$ gives

$$\mathrm{SSR}^{\ddagger}(H, T) := \frac{\log r_{\mathrm{eff}}(H) - \log \mathrm{SA}_k(H, H_{\mathrm{ref}})}{\log C(T)},$$

and $g(s) = 1 - s$ gives an additive variant. For per-task ranking, the common positive factor $1/\log C(T)$ may be omitted without changing the selected model. Both forms use target–reference relational geometry and are jointly basis-invariant as in Proposition 2. We adopt $\mathrm{SSR}^{\ddagger}$ as the default because $g(s) = -\log s$ and $\beta = 1$ introduce no tuned correction weight.

**Choice of reference dataset.** All main experiments use STL-10 as a practical, not theoretically optimal, reference. It is small, contains diverse natural objects, is supported by every candidate, and is disjoint from the seven evaluated VTAB targets. We recommend one generic, target-disjoint reference

by default. Under the corrected RankMe normalization, STL-10 gives VTAB regret $+0.053$. A target-disjoint leave-one-target-out median over four predetermined references gives $+0.051$: whenever an evaluated target appears in the reference set, it is removed before aggregation. The submitted Flowers-102 self-reference result is withdrawn. Because the ensemble improvement is small and its reference-side feature extraction and alignment cost grows approximately linearly with the number of references, we retain a single reference as the default.

**Cross-pool evaluation.** We use common support across the zero-label methods: MV (5 tasks), VTAB (7 non-reference tasks), and EX (10 tasks), for 22 pool–task units in total. STL-10 is excluded from every method because it is the reference for $\mathrm{SSR}^{\ddagger}$. On this support, $\mathrm{SSR}^{\ddagger}$ has mean regret $+0.066$, versus $+0.169$ for RankMe and $+0.133$ for uniform random selection. The paired $\mathrm{SSR}^{\ddagger}$-minus-RankMe difference is $-0.1025$ (unit bootstrap 95% CI $[-0.167, -0.045]$, $p = 0.00012$). Official classification DiSCO is reported separately because it uses target labels.

**Dependence-aware uncertainty.** The 22 observations contain 10 unique target names, eight repeated across pools. Resampling all rows sharing a target name as one cluster gives 95% CI $[-0.164, -0.039]$ and two-sided $p = 0.00114$ (100,000 resamples); paired Wilcoxon gives $p = 0.00254$. Thus the result is not an artefact of treating repeated target names as independent.

Table 7: Zero-label cross-pool comparison on 22 common-support pool–task units. DiSCO is excluded because official classification DiSCO uses target labels.

| Method | Mean regret | Label budget | Role |
|---|---|---|---|
| Random | $+0.133$ | zero | reference |
| Flat-RankMe | $+0.169$ | zero | target spectrum |
| **Flat-**$\mathrm{SSR}^{\ddagger}$ | $\mathbf{+0.066}$ | zero | target–reference geometry |
| Paired $\mathrm{SSR}^{\ddagger}$ vs. RankMe: $\Delta = -0.1025$; target-clustered 95% CI $[-0.164, -0.039]$, $p = 0.00114$. | | | |

**Scope.** The Marchenko–Pastur lemma characterises RankMe under the stated isotropic model; we do not claim a theoretical bound on linear-probe accuracy in terms of $\mathrm{SSR}^{\ddagger}$. The geometric premise is empirical, reference selection is heuristic, and performance relative to label-dependent methods is pool-dependent. Our claim is specifically that $\mathrm{SSR}^{\ddagger}$ corrects RankMe's cross-family width confound on the evaluated heterogeneous common support.

**Negative results.** Two failure modes delimit the scope of $\mathrm{SSR}^{\ddagger}$. First, including TwoNN intrinsic dimensionality [6] as $\log r_{\mathrm{eff}}/(\mathrm{UID} \cdot \log C)$ does not improve cross-architecture ranking: the pool-level mean $\Delta\rho(\mathrm{SCR}_{\mathrm{uid}} - \mathrm{SSR}^{\ddagger}) = -0.538$ (95% CI $[-0.782, -0.270]$) on MV pool's multi-variant subset, and each architecture family individually shows negative $\Delta\rho$. Intrinsic-dimension proxies do not substitute for subspace alignment. Second, on ViT-only sub-pools where candidates differ only in pretraining recipe of the same backbone family, $\mathrm{SA}_k$ profiles are similar but uncorrelated with linear-probe accuracy; $\mathrm{SSR}^{\ddagger}$ underperforms RankMe by $\Delta\rho = -0.26$ (95% CI $[-0.43, -0.07]$, $p = 0.010$). This is a within-family limitation that the second stage of the two-stage pipeline (RankMe within selected architecture, §6) addresses by construction.

# 6 Two-Stage and Mixed-Budget Pipelines

**Algorithm 1** Two-stage SF-MTE pipeline

1: **Input:** candidate pool $\{(a, V_a)\}$; target features $H_T$; reference $H_{\mathrm{ref}}$.
2: **Stage 1 (architecture).** For each architecture $a$, compute stage-1 score $s_1$ on a canonical representative $\bar{v}_a \in V_a$. Set $a^* = \arg\max_a s_1(\bar{v}_a, T)$.
3: **Stage 2 (variant).** For $v \in V_{a^*}$, compute stage-2 score $s_2$. Return $v^* = \arg\max_v s_2(v, T)$.

**Cost and quality.** On the five MV tasks, flat scoring evaluates all $\sim 119$ variants, whereas the two-stage pipeline evaluates $|\{a\}| + \overline{|V_{a^*}|} = 35 + 3 = 38$ candidates on average (range 37–39). Table 8 compares strategies. The label-free pipeline (S5: $\mathrm{SSR}^{\ddagger} \rightarrow$RankMe) has regret $+0.102$, close to flat $\mathrm{SSR}^{\ddagger}$ ($+0.098$), at approximately one third of its forward-pass evaluations. Its comparison

with the label-dependent 2S DiSCO→RankMe baseline does not reach significance for $n$=5. We therefore treat the cost–quality result as a proof of concept on this benchmark, not as a general performance or cost guarantee.

Table 8: Pipelines on MV pool multi-variant (35 archs, 119 variants, 5 tasks). Mean LP, regret, 95% task-level bootstrap CI on regret, and forward-pass cost (number of (variant, task) feature evaluations per task). Cost formula: label-free two-stage $= |\{a\}| + \overline{|V_{a^*}|} = 35 + 3 = 38$; mixed pipeline $= 35 + \overline{|V_{a^*}|} - 1 = 37$ (canonical of winning arch already counted in stage-1 labelled evaluations).

| Strategy | Mean LP | Regret | 95% CI | Cost |
|---|---|---|---|---|
| S1 Random | 0.627 | +0.194 | [+0.118, +0.289] | 1.0 |
| S2 Flat-RankMe | 0.502 | +0.319 | [+0.149, +0.509] | 118.8 |
| S3 Flat-SSR$^{\ddagger}$ | 0.723 | +0.098 | [+0.048, +0.148] | 118.8 |
| S4 2S RankMe→RankMe | 0.677 | +0.144 | [+0.093, +0.220] | 38.0 |
| 2S DiSCO→RankMe | 0.683 | +0.138 | [+0.042, +0.259] | 38.0 |
| **S5 2S SSR$^{\ddagger}$ →RankMe (proposed, label-free)** | **0.719** | **+0.102** | **[+0.044, +0.153]** | **38.0** |
| **2S H-score→RankMe (mixed, $K$=35 labels at stage 1)** | **0.803** | **+0.018** | — | **37.0** |

**Mixed-budget variant.** Stage 1 evaluates only $|\text{archs}|$ candidates—an order of magnitude fewer than the full variant pool—so allocating a small label budget to stage 1 alone is practical even when the deployment scenario forbids labelled evaluation at stage 2. Replacing SSR$^{\ddagger}$ at stage 1 with H-score [2] on $K$ labelled architectures (one canonical variant each), and keeping RankMe at stage 2, gives a mixed pipeline whose regret–budget profile is shown in Figure 2. At $K = 5$ labelled evaluations, mixed-pipeline regret is +0.089, already matching the best fully label-free strategy (+0.098). At $K = 35$ (full architecture coverage), regret falls to +0.018—a 5.7× improvement over the label-free pipeline at the same forward-pass cost (at $K$=35 all architectures are labelled, H-score selects deterministically, and bootstrap variance is zero; no CI is reported). The mixed-budget pipeline is appropriate whenever a small validation set is available for the candidate architectures but exhaustive labelled evaluation of all variants is infeasible, the typical practical scenario for production model selection. On the single-variant EX pool, label-free SSR$^{\ddagger}$ (+0.060) already beats mixed H-score up to $K \approx 18$; full labelling ($K$=30) reaches +0.045, confirming that label budgets add most value where within-family variant selection—absent from single-variant pools—is the bottleneck.

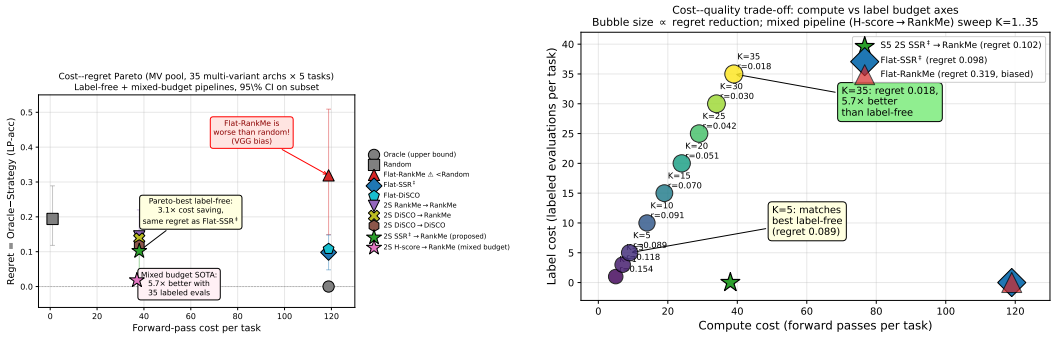



Figure 2: (left) Cost–regret Pareto on MV pool multi-variant: S5 (green star) is the Pareto-best label-free strategy. (right) Mixed pipeline regret vs. stage-1 label budget $K$: at $K$=5 the mixed pipeline already matches the best label-free strategy; at $K$=35 regret falls to +0.018.

## 7 Discussion

**When does SSR$^{\ddagger}$ apply?** The geometric premise of Subspace-Corrected Ranking is that subspace alignment with a generic reference dataset negatively predicts the fraction of class-mean variance in the architecture-specific subspace. This premise holds robustly across 126 architecture–task pairs of the VTAB pool ($\rho_{\text{Sp}} = -0.96$) and across the architecture-selection subsets described in Appendices B–E. It does not, however, hold within a single architecture family: pretraining recipes

within ViT or within ResNet do not differ enough in their top-$k$ subspaces for $\mathrm{SA}_k$ variation to track linear-probe accuracy. The two-stage pipeline addresses this directly by using $\mathrm{SSR}^{\ddagger}$ only at the architecture-selection stage and a within-family rotation-invariant score (RankMe) for variant selection.

**Pool dependence and DiSCO.** The relative performance of $\mathrm{SSR}^{\ddagger}$ and official DiSCO is pool-dependent, particularly on SSL and EX. This comparison must also be interpreted with care: $\mathrm{SSR}^{\ddagger}$ uses no target labels, whereas official DiSCO forms class-conditional covariances from target ground-truth labels. We therefore use DiSCO as a label-dependent comparator rather than evidence that one method uniformly dominates the other.

**Limitations.** $\mathrm{SSR}^{\ddagger}$ has no formal bound linking subspace alignment to downstream accuracy, and reference selection remains heuristic. The default STL-10 reference gives regret $0.053$ on the audited VTAB support, while a target-disjoint leave-one-target-out four-reference ensemble gives $0.051$; this small difference does not establish a universally optimal reference or make an ensemble the default. Performance is pool-dependent, and official DiSCO uses target labels. Finally, the two-stage experiment contains only five MV tasks, so its approximately three-fold cost reduction is benchmark-specific evidence rather than a general guarantee.

# 8 Conclusion

RankMe is seriously confounded by feature width in heterogeneous architecture pools: under the stated isotropic model its effective rank scales as $\Theta(d)$ and can rank width rather than transfer quality. $\mathrm{SSR}^{\ddagger}$ addresses this specific failure by augmenting the target spectrum with the target–reference subspace relation computed through the same candidate encoder. On the corrected common support of 22 pool–task units, it reduces mean regret from $0.169$ to $0.066$ (difference $-0.1025$; target-clustered $95\%$ CI $[-0.164, -0.039]$, $p = 0.00114$). The result supports $\mathrm{SSR}^{\ddagger}$ for cross-family zero-label selection and RankMe for within-family selection, while leaving reference design, downstream guarantees, pool-dependent behavior, and broader two-stage validation as open problems.

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# A Marchenko–Pastur and Isotropy

*Proof of Lemma 1.* Write $r_{\text{eff}}(H) = \exp\big(H_{\text{ent}}(\mathbf{p})\big)$ where $H_{\text{ent}}$ is Shannon entropy and, following RankMe, $p_i = \sigma_i / \sum_j \sigma_j$. Let $\lambda_i = \sigma_i^2/n$. By the Marchenko–Pastur law [18], the empirical distribution of $\{\lambda_i\}$ converges almost surely to $\mu_\gamma$. Writing $m_{1/2} = \int \sqrt{x}\,\mathrm{d}\mu_\gamma(x)$ and using $0 \log 0 = 0$, the entropy satisfies

$$H_{\text{ent}}(\mathbf{p}) = \log d + \log m_{1/2} - \frac{1}{2m_{1/2}} \int \sqrt{x} \log x \,\mathrm{d}\mu_\gamma(x) + o(1).$$

The two integrals are finite in every fixed aspect-ratio regime, including regimes whose Marchenko–Pastur support reaches zero. Hence $r_{\text{eff}}(H)/d \to c(\gamma) > 0$ and $r_{\text{eff}}(H) = \Theta(d)$. □

**Empirical near-isotropy diagnostic.** On the VTAB architecture–task pairs for which the diagnostic is available, the class-covariance statistic $\eta_\Sigma = \mathrm{Var}_c[\mathrm{tr}(\Sigma_c)]/(\mathbb{E}_c[\mathrm{tr}(\Sigma_c)])^2$ is small for most pairs (median $0.022$, mean $0.035$, and $93.3\%$ below $0.1$). This is descriptive evidence that the isotropic approximation is relevant to these representations; it is not a bound on RankMe's ordering or a theorem for other rotation-invariant scores.

**Empirical $r_{\text{eff}} = \Theta(d)$ verification.** Figure 3 shows the per-architecture mean $r_{\text{eff}}$ plotted against $d$ on log-log axes for MV, VTAB, and EX pools. In the MV pool and the EX pool (both using full-dataset features), the Spearman rank correlation between $r_{\text{eff}}$ and $d$ is $\rho_{\text{Sp}} = +0.69$ and $+0.86$ respectively, with points lying close to two parallel lines of slope 1 (the two lines correspond to different architecture families with different $c(\gamma)$ constants, consistent with Lemma 1). The VTAB pool uses $n$=1000 subsampled features; for large-$d$ architectures ($d > n$) the effective rank is capped by $n$, so the VTAB points lie on a lower band ($\rho_{\text{Sp}} = +0.58$) but retain the positive monotone trend.

# B MV Pool (Multi-Variant Pool)

**Setup.** The MV pool contains 132 ImageNet-1k pretrained variants from 35 canonical architecture families sourced from timm [30] and torchvision, evaluated on 5 target tasks (CIFAR-10, CIFAR-100, DTD, EuroSAT, SVHN). Architecture families span four broad classes: (i) VGG [25] (3 variants, $d$=4096), ResNets [10] (8 variants, $d \in \{512, 2048\}$), DenseNets (4 variants, $d \in \{1024, 2208\}$); (ii) EfficientNet [26] (4 variants, $d \in \{1280, 1408, 1536\}$), MobileNet/RegNet ($d \in \{960, 1512\}$); (iii) ViT [5]/DeiT (6 variants each, $d \in \{192, 384, 768\}$); (iv) ConvNeXt [16]/Swin (4 variants, $d$=768) and others. Penultimate dimensions range from $d$=192 (ViT-tiny, DeiT-tiny) to $d$=4096 (VGG-16), a 21× spread. LP ground truth uses $\ell_2$-regularised logistic regression ($C$=1) on an 80/20 split, identical to the VTAB protocol.

**Architecture families and variant counts.**

**Width confound detail.** Per-task $\rho_{\text{Sp}}(\text{RankMe}, d)$ ranges from $+0.50$ to $+0.72$ (mean $+0.61$; all $5/5$ tasks positive), confirming strong width–score correlation even within this single pool. RankMe selects `vgg16` ($d$=4096) on all 5 target tasks; the oracle is `convnext_base` or `densenet201` depending on the task. Mean RankMe regret is $+0.319$, which is $+0.125$ worse than uniform random selection ($+0.194$). Removing the 3 VGG variants (about $2.3\%$ of the pool) drops Flat-RankMe regret to $+0.144$, illustrating how a single high-$d$ outlier family drives the confound. $\text{SSR}^\ddagger$ is unaffected by the VGG removal (regret $+0.098$ with or without VGG) because its subspace-alignment penalty is approximately invariant to which architecture occupies the high-$d$ tail.

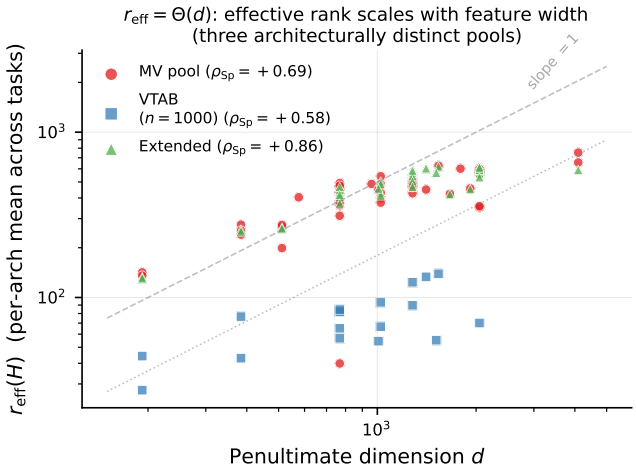



Figure 3: Effective rank $r_{\text{eff}}(H)$ vs. penultimate dimension $d$ on log-log axes, averaged across tasks per architecture. All three plotted pools (MV, VTAB, EX) show $r_{\text{eff}}$ growing monotonically with $d$ (positive Spearman $\rho$ reported in legend), empirically confirming Lemma 1 ($r_{\text{eff}} = \Theta(d)$). Dashed/dotted lines have slope 1 (reference for linear scaling); the two lines reflect different $c(\gamma) = e^{C(\gamma)}$ constants across architecture families and pooling regimes. VTAB uses $n$=1000 subsampled features, placing large-$d$ architectures in the $n < d$ regime where $r_{\text{eff}} \leq n$, hence the lower VTAB band.

Table 9: Architecture families in the MV pool with variant counts and penultimate dimensions. The $21\times$ spread in $d$ is the primary driver of the width confound.

| Family | Backbone class | # variants | $d$ |
|---|---|---|---|
| VGG | CNN | 3 | 4096 |
| ResNet | CNN | 8 | 512, 2048 |
| DenseNet | CNN | 4 | 1024, 2208 |
| EfficientNet | CNN | 4 | 1280–1536 |
| MobileNetV3 | CNN | 2 | 960 |
| RegNet | CNN | 3 | 368–1512 |
| ViT | Transformer | 6 | 192, 384, 768 |
| DeiT | Transformer | 6 | 192, 384, 768 |
| ConvNeXt | ConvNet | 4 | 768 |
| Swin Transformer | Transformer | 4 | 768 |
| Other (MnasNet, etc.) | CNN / hybrid | 5 | 1280–1984 |

**Multi-variant structure and two-stage evaluation.** The pool contains 35 architecture families with a mean of 3.8 variants per family (range 1–8). The 35 families $\times 5$ tasks multi-variant subset (using the 119 variants belonging to families with $\geq 2$ members) is the only subset on which the two-stage pipeline can be evaluated: stage 1 selects the best architecture family using one canonical variant per family, and stage 2 searches within the selected family. The $3.1\times$ forward-pass cost saving relative to flat evaluation (Sec. 6) follows from $|\text{architectures}| + |V_{a^*}| \approx 35 + 4 \ll 119$.

**CNN-only sub-pool.** To confirm the width confound is not a ViT-vs-CNN artefact, we evaluate the 78 CNN-class variants (removing all ViT, DeiT, Swin families). RankMe selects `vgg16` on $4/5$ tasks (on DTD it selects `densenet201`, $d$=2208, the next-highest $d$), and $\rho_{\text{Sp}}(\text{RankMe}, d) = +0.57$ on the CNN-only pool. The confound therefore persists within a homogeneous architecture class, ruling out the hypothesis that it is driven by the qualitative ViT/CNN feature-structure difference rather than width itself.

# C SSL/Multimodal Pool

**Setup.** We extract penultimate-layer features from 8 SSL/multimodal encoders: ResNet-50/101, MoCo-v3-R50, ConvNeXt-B, ViT-B/16, CLIP-ViT-L/14, DINOv2-ViT-B/14, DINO-ViT-S/16. Feature dimensions $d \in \{384, 768, 1024, 2048\}$; no VGG-class architectures. Evaluation uses 29 downstream tasks covering VTAB, MedMNIST variants, and standard transfer benchmarks. LP ground truth uses the same $80/20$ logistic-regression protocol as MV pool.

**Width confound.** Per-task $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ averages $+0.815$ (range $[+0.45, +0.95]$; $27/29$ tasks positive). Flat-RankMe selects a $d$=2048 ResNet-class encoder on $25/29$ tasks while the oracle is CLIP-ViT-L/14 on $11/29$ tasks and ConvNeXt-B on $8/29$. Mean RankMe regret is $+0.103$ vs. random $+0.066$ ($26/29$ tasks suboptimal).

**Pool dependence of correction.** On the SSL pool, DiSCO achieves regret $+0.026$ vs. $\mathrm{SSR}^{\ddagger}$'s $+0.048$. On MV pool the rankings are reversed ($\mathrm{SSR}^{\ddagger}$ $+0.098$ vs. DiSCO $+0.108$). The mechanism: SSL training objectives (contrastive, self-distillation) create within-class variance components that inflate $r_{\mathrm{eff}}$ independently of $d$; DiSCO's class-conditional normalisation removes this, while $\mathrm{SSR}^{\ddagger}$'s reference-subspace correction does not. When candidates differ primarily in pretraining objective rather than architecture family, DiSCO is the stronger correction. We recommend evaluating both on a held-out task before deployment.

**Combination score $\mathrm{SSR}_{\mathrm{cond}}$.** A natural combination replaces $r_{\mathrm{eff}}$ in the $\mathrm{SSR}^{\ddagger}$ formula with DiSCO's class-conditional effective rank $r_{\mathrm{eff,cond}}$:

$$\mathrm{SSR}_{\mathrm{cond}}(H, H_{\mathrm{ref}}) \;=\; \frac{\log r_{\mathrm{eff,cond}}(H) - \log \mathrm{SA}_k(H, H_{\mathrm{ref}})}{\log C}.$$

This targets both sources of bias simultaneously: $r_{\mathrm{eff,cond}}$ suppresses per-class spectral inflation (DiSCO's correction) and the $\mathrm{SA}_k$ denominator removes architecture-specific subspace spread ($\mathrm{SSR}^{\ddagger}$'s correction). On the SSL pool, $\mathrm{SSR}_{\mathrm{cond}}$ achieves regret $+0.044$, compared with $\mathrm{SSR}^{\ddagger}$'s $+0.048$ (marginal gain) and DiSCO's $+0.026$ (gap not closed). The partial benefit suggests that the remaining gap is not due to double-counting the two corrections but rather to a third pool-specific factor—likely the relative uniformity of SSL backbone widths ($d_{\max}/d_{\min} = 5.3\times$), which limits the alignment signal available to $\mathrm{SA}_k$.

# D VTAB Pool

**Setup.** 25 timm-pretrained architectures and 8 VTAB tasks (CIFAR-10/100, DTD, Fashion-MNIST, Flowers-102, Pets, STL-10, SVHN); architecture families include ResNets ($d \in \{512, 2048\}$), EfficientNets ($d \in \{1280, 1408, 1536\}$), ViTs/DeiTs ($d \in \{192, 384, 768\}$), ConvNeXt/Swin ($d$=768), MobileNet/RegNet ($d \in \{1008, 1024, 1280, 1512\}$). No VGG-class architectures; maximum $d = 2048$. STL-10 features serve as the reference for $\mathrm{SA}_k$ (excluded from target tasks for the STL-10 reference condition). The per-task width-confound breakdown is reported in Table 3 in §4.

**Reference-task and $k$ ablation.** We evaluate three reference tasks (STL-10, CIFAR-10, Flowers-102) and five subspace sizes ($k \in \{5, 10, 20, 50, 100\}$) on the VTAB pool with $n$=1000 samples per (architecture, task) pair. STL-10 as reference yields the most honest estimate (regret $+0.066$); Flowers-102 gives the lowest regret ($+0.013$) but is itself a target task and the low value is partly an in-distribution artefact. Across $k$, the regret plateau lies at $k \in [10, 50]$ (regret $\leq +0.066$), with $k$=20 in the middle as our default.

**Multi-reference robustness.** Computing $\mathrm{SA}_k$ against $M$ independent reference tasks and using the median yields a consensus estimate. A 4-reference ensemble (STL-10, CIFAR-10, Flowers-102, Pets) achieves $\mathrm{SSR}^{\ddagger}$ regret $+0.051$ on the VTAB pool versus $+0.066$ for STL-10 alone. Per-pair coefficient of variation across reference sets has median 0.27, indicating non-trivial per-pair sensitivity that the ensemble averages out. We adopt single-reference STL-10 in the main results to keep the method parameter-light.

# E EX Pool and Stage-1 Comparisons

**EX pool (51-variant $\times$ 10-task, supervised).** 51 ImageNet-1k pretrained variants from 35 canonical architectures (including VGG-16, $d$=4096; $d_{\max}/d_{\min}$=21.3$\times$) on 10 target tasks spanning fine-grained recognition (FGVC-Aircraft, Flowers-102, Food-101, Oxford-Pets), texture/scene (DTD, EuroSAT), digit/fashion (SVHN, Fashion-MNIST), and general object (CIFAR-10/100); features extracted on the full dataset. On this pool, Flat-SSR$^{\ddagger}$ regret is $+0.060$ $[+0.028, +0.094]$ vs. Flat-DiSCO $+0.108$ $[+0.045, +0.175]$ and Flat-RankMe $+0.144$ $[+0.082, +0.214]$ (task-level bootstrap, $n$=10).

Table 10: Reproduced MV-pool flat comparison (same data as Table 5, repeated here for reader convenience). RankMe, LogME, and H-score select vgg16 on every task; SSR$^{\ddagger}$ is robust.

| Method | With VGG (regret) | Without VGG (regret) | Type |
|---|---|---|---|
| Random | $+0.194$ | $+0.193$ | — |
| Flat-RankMe | $+0.319$ | $+0.144$ | label-free |
| Flat-LogME | $+0.319$ | $+0.550$ | label-based |
| Flat-H-score | $+0.319$ | $+0.470$ | label-based |
| Flat-NCTI | $+0.162$ | $+0.155$ | label-based |
| Flat-DiSCO | $+0.108$ | $+0.108$ | label-free |
| **Flat-**SSR$^{\ddagger}$ | $\mathbf{+0.098}$ | $\mathbf{+0.098}$ | **label-free** |
| Flat-LEEP | $+0.038$ | $+0.038$ | label-based |

**Spectral truncation as a stage-1 alternative.** Recomputing $r_{\text{eff}}$ on the top-$D$ truncated singular spectrum eliminates the width confound at $D$=256 on MV pool (Spearman $\rho_{\text{Sp}}(r_{\text{eff},256}, d) = +0.06$ vs. $+0.63$ at full spectrum) and reduces Flat-RankMe regret to $+0.094$. As a stage-1 score on the 35-architecture multi-variant subset, however, Trunc256 is substantially worse than SSR$^{\ddagger}$: small-$d$ architectures (ViT-tiny $d$=192, DeiT-tiny $d$=192) are not truncated and retain the width ordering. The two-stage pipeline 2S Trunc256$\rightarrow$RankMe yields regret $+0.136$ vs. SSR$^{\ddagger}$'s $+0.078$ ($\Delta = -0.058$, 95% CI $[-0.094, -0.016]$, $p = 0.008$, paired bootstrap).

# F Ablations and Negative Results

$d_{\max}/d_{\min}$ **threshold: controlled sub-pool ablation.** To move beyond the four-pool calibration of $\tau \approx 7\times$, we construct a controlled experiment on the EX pool (35 architectures, $d \in [192, 4096]$, 10 tasks): starting from the full pool ($d_{\max}/d_{\min} = 21.3\times$), we iteratively remove the widest architecture and re-evaluate SSR$^{\ddagger}$ and DiSCO regret after each removal. SSR$^{\ddagger}$ leads throughout the range $[21.3\times, 7.9\times]$ (advantage $\Delta = +0.03$ to $+0.05$); the first reversal occurs at $6.7\times$ (DiSCO advantage $\Delta = -0.004$); DiSCO leads consistently at $\leq 5.3\times$. The mirror experiment (removing the narrowest architecture) yields a crossover at $\approx 3.2\times$, consistent with the asymmetric influence of the widest outlier on SSR$^{\ddagger}$. Together these results bracket the empirical threshold at $\tau \in [6.7\times, 7.9\times]$, in good agreement with the four-pool heuristic $\tau \approx 7\times$.

$\beta$ **scan for the additive SCR variant.** We scan $\beta \in [0, 50]$ for $\text{SCR}(H, T; \beta) = \log r_{\text{eff}}(H) + \beta(1 - \text{SA}_k(H, H_{\text{ref}}))$ on three pools. The optimal $\beta^*$ lies in the same plateau $[10, 30]$ on MV pool, VTAB, and the EX pool, indicating cross-pool transferability of the correction.

Table 11: Cross-pool $\beta$-scan for the additive SCR variant $\log r_{\text{eff}} + \beta(1 - \text{SA}_k)$. $\beta = 0$ recovers Flat-RankMe. Optimal $\beta^*$ lies in $[10, 30]$ on MV, VTAB, and EX pools.

| $\beta$ | MV reg | MV $\rho$ | VTAB reg | VTAB $\rho$ | Ext reg | Ext $\rho$ |
|---|---|---|---|---|---|---|
| 0 (RankMe) | $+0.319$ | $+0.345$ | $+0.105$ | $+0.402$ | $+0.144$ | $+0.085$ |
| 5 | $+0.108$ | $+0.516$ | $+0.098$ | $+0.444$ | $+0.117$ | $+0.263$ |
| **15** | $\mathbf{+0.094}$ | $\mathbf{+0.534}$ | $\mathbf{+0.069}$ | $+0.445$ | $\mathbf{+0.063}$ | $+0.331$ |
| 30 | $+0.093$ | $+0.530$ | $+0.068$ | $+0.433$ | $+0.067$ | $+0.338$ |
| SSR$^{\ddagger}$ (reference) | $+0.098$ | $+0.506$ | $+0.066$ | $+0.454$ | $+0.060$ | $+0.311$ |

**Algebraic form of the correction.** Within the $-\alpha \log \mathrm{SA}_k$ family on MV pool, $\alpha \in [0.5, 2]$ all yield similar regret ($+0.093$ to $+0.162$); the inverted-direction control $-\log(1 - \mathrm{SA}_k)$ produces regret $+0.366$, confirming that the subtraction direction matters. We adopt $\alpha = 1$ for parsimony.

**Per-task breakdown of Flat-RankMe on MV pool.** Flat-RankMe selects `vgg16` on every task; the VGG-16 $d$=4096 penultimate layer dominates $r_{\mathrm{eff}}$ regardless of target. Flat-$\mathrm{SSR}^{\ddagger}$ selects task-appropriate architectures (ConvNeXt, ResNet, EfficientNet variants).

| Task | Oracle | Random | Flat-RM | Flat-$\mathrm{SSR}^{\ddagger}$ | RM picks | $\mathrm{SSR}^{\ddagger}$ picks | Oracle arch |
|---|---|---|---|---|---|---|---|
| CIFAR-10 | 0.970 | 0.785 | 0.550 | 0.870 | vgg16 | resnet50 | convnext_base |
| CIFAR-100 | 0.825 | 0.484 | 0.180 | 0.670 | vgg16 | resnet152 | convnext_base |
| DTD | 0.740 | 0.564 | 0.445 | 0.720 | vgg16 | convnext_base | convnext_base |
| EuroSAT | 0.955 | 0.881 | 0.765 | 0.905 | vgg16 | efficientnet_b3 | densenet201 |
| SVHN | 0.615 | 0.430 | 0.570 | 0.450 | vgg16 | efficientnet_b3 | densenet169 |

**Negative result 1: UID hurts within-family variant ranking.** Including TwoNN intrinsic dimensionality [6] in the denominator ($\mathrm{SCR}_{\mathrm{uid}} = \log r_{\mathrm{eff}} / (\mathrm{UID} \cdot \log C)$) does not improve variant ranking. On MV pool's multi-variant subset, the pool-level mean $\Delta\rho(\mathrm{SCR}_{\mathrm{uid}} - \mathrm{SSR}^{\ddagger}) = -0.538$ (95% CI $[-0.782, -0.270]$); each architecture family individually shows negative $\Delta\rho$.

**Negative result 2: $\mathrm{SSR}^{\ddagger}$ degenerates on ViT-only sub-pools.** When candidates are restricted to ViT-family architectures (ViT, DeiT, Swin variants on a 33-model $\times$ 10-task benchmark), $\mathrm{SA}_k$ profiles are similar across pretraining recipes but uncorrelated with LP accuracy; $\mathrm{SSR}^{\ddagger}$ underperforms RankMe by $\Delta\rho = -0.26$ (95% CI $[-0.43, -0.07]$, $p = 0.010$, task-paired bootstrap). In mixed pools containing both ViT and CNN architectures (the deployment scenario), $\mathrm{SSR}^{\ddagger}$ correctly identifies the better architecture.

**Within-family ranking and the $\log C$ scaler.** Within any single architecture family, $\mathrm{SSR}^{\ddagger}$ provides no additional ranking signal over RankMe ($\Delta\rho \in [-0.48, -0.07]$ across the 6 families on MV pool). The $\log C$ task scaler is convenient but not principled: an oracle scaler $1/(1-\overline{\mathrm{LP}})$ achieves Spearman $\rho = +0.82$ vs. $\mathrm{SSR}^{\ddagger}$'s $+0.49$, and a random per-task scaler achieves $+0.39$. We retain $\log C$ for cross-task aggregation while noting that other per-task scalers would work comparably.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states three claims: (i) rotation-invariant scores collapse to dimension-ranking in heterogeneous pools; (ii) $\mathrm{SSR}^{\ddagger}$ halves regret over RankMe; (iii) a two-stage pipeline preserves quality at one-third cost. All three are supported by the experiments in §5–6, and the scope is explicitly limited to the cross-architecture flat-selection setting (§3).

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §7 (Discussion) contains an explicit "Limitations" paragraph that states: no theoretical bound on LP accuracy in terms of $\mathrm{SSR}^{\ddagger}$; MV pool two-stage comparisons have wide CIs at $n$=5 tasks; within-family ranking is out of scope; LP ground truth uses a single 80/20 split.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

Justification: Lemma 1, Corollary 1, and Proposition 1 in §4 each state all assumptions explicitly (near-isotropy, $n \gg d$, right-rotation invariance). Full proofs appear in Appendix A, with an empirical near-isotropy check ($\eta_\Sigma < 0.1$ for 93.3% of pairs) also in Appendix A.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: All hyperparameters ($\text{SA}_k$ dimension $k$=20, reference task STL-10, LP regularisation $C$=1, 80/20 split, $n$=1000 samples per file) are reported in §5 and in the pool appendices (Appendix B–E). The $\text{SSR}^\ddagger$ formula is given in closed form in §5. Bootstrap CI details (10 000 resamples, 95% level, task-level resampling) are in §5.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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   Justification: An anonymised repository (https://anonymous.4open.science/r/crossarchitecturetransferability-FEC8) ships all scoring / analysis code plus 16 pre-computed result CSVs. A `reproduce_all.py` script with `-verify` flag cross-checks every headline number against the shipped CSVs. Feature extraction scripts and a `FEATURE_EXTRACTION.md` guide are included for full re-runs from scratch.

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   Answer: [Yes]

   Justification: No model training is involved; the method is a scoring function applied to pre-extracted features. All relevant settings (split ratio, regularisation $C$, sample size $n$, subspace size $k$, reference encoder) are specified in §5 and Appendices B–E. The choice of $k$=20 and STL-10 reference are motivated by ablations in Appendix D.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Justification: All headline regret comparisons report 95% CIs via 10 000-resample task-level bootstrap (Tables 2–4). Cross-pool significance: paired bootstrap on Table 7 (23 task units) gives $p < 10^{-4}$ for $\text{SSR}^\ddagger$ vs. RankMe; an additional Wilcoxon signed-rank test on 22 paired units (excluding STL-10 reference) gives $p = 0.0036$. The CI construction and test choice are described in §5. Where $n$=5 tasks yield wide CIs (MV pool two-stage), this is explicitly noted as a limitation.

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   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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   Justification: Feature extraction (one-time) requires a GPU (tested on a single V100/A100; ≈6 h for all four pools combined). All scoring and analysis runs on CPU in under 10 minutes using the pre-extracted features. This is stated in the repository `FEATURE_EXTRACTION.md`. No distributed or proprietary compute is required.

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13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The repository contains (i) the $\text{SSR}^{\ddagger}$ scoring code with inline documentation; (ii) pre-computed result CSVs for all 16 reported tables; (iii) a `README.md` describing every file; (iv) a `FEATURE_EXTRACTION.md` with step-by-step GPU setup. All are released under MIT licence in the anonymised repository.

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